

Integer Pressure and a Missing Taylor Coefficient

Analytic phase dependence and the sixth-moment optimum for generalized Thue–Morse products

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Abstract

For the doubling map and the translated potential $\psi_c(x) = \log \cos^2 \pi(x - c)$, we give an explicit finite-dimensional model of every positive integer pressure $p_m(c) = P(m\psi_c)$. Its Perron eigenvalue is algebraically simple for every phase, including the atomic phase $c = 0$, and is uniformly separated from the remaining spectrum when m is fixed. Consequently, every function $c \mapsto p_m(c)$ is real-analytic. This extends the previously established order-two phase regularity to all positive integer orders; it does not settle noninteger orders.

The finite model also reveals an all-order cancellation: $[c^{2m}]p_m(c) = 0$. All lower even coefficients are explicit zeta values. The mechanism is delayed feedback from the second inverse branch, whose weight first appears in degree $2m$, together with an exact periodized-sinc Perron eigenfunction at $c = 0$.

A separate exact optimization result shows that the classical Thue–Morse phase $c = 1/2$ minimizes the fourth-moment growth rate but is a strict local maximum for the sixth-moment rate. We characterize the two global sixth-moment minimizers by a degree-fourteen polynomial and certify their uniqueness using rational Sturm and interval arithmetic. One minimizer is approximately 0.364919195551255.

The article supplies proofs, explicit matrices, recurrence formulas, reproducible exact-arithmetic checks, a connection to the Fabius/Rvachev Fourier program, and a proposed research agenda. The new claims are research results argued here, not independently refereed conclusions or Lean-verified theorems. Prior moment-matrix methods are explicitly credited.

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1 The research question and the contribution

1.1 A concrete question in the literature

The family of singular potentials

$$T(x) = 2x \pmod{1}, \quad g_c(x) = \cos^2 \pi(x - c), \quad \psi_c = \log g_c$$

is a natural deformation of the classical Thue–Morse sine product. Gohlke, Kesseböhmer, and Schindler establish real-analytic dependence on c for the pressure at order two and explicitly leave stronger regularity at general positive orders open [1, Theorem 2.7 and following discussion]. Their pressure–spectrum identity also connects the question to generalized Thue–Morse diffraction measures [1, Theorem 2.6].

Here the integer cases admit a particularly effective answer. Although ψ_c has a logarithmic singularity, $\exp(m\psi_c) = g_c^m$ is a trigonometric polynomial when m is a positive integer. The associated transfer operator therefore has a small invariant space. The substantive point is not merely writing down this space: a complete argument must identify its spectral radius with pressure, prove simplicity at every phase, and handle the exceptional phase $c = 0$ without silently assuming irreducibility there.

1.2 Relationship to the repository

The inspected ProveIt snapshot contains a large Fabius/Rvachev development and a Fourier-decay synthesis [6, 7]. The synthesis already develops lacunary sine cocycles, pressure-based moment exponents, and several distinct interpretations of Fourier decay. These are not reintroduced as new results here. We instead study phase dependence and phase optimization of the finite cocycle underlying that program.

Our arguments are self-contained at the level of finite-dimensional spectral theory and elementary Fourier analysis. They do not assume that a claim is correct merely because it appears in a repository draft. In particular, none of the conclusions below requires a general infinite-dimensional Ruelle–Perron–Frobenius theorem for a singular potential. The measure-theoretic corollaries are explicitly separated and uses the pressure–spectrum theorem of [1].

1.3 What is old, what is proved here, and what remains open

Finite recurrences for the classical even moments are established work. Mauduit, Montgomery, and Rivat obtained an order- m recurrence and an $m \times m$ matrix for the $2m$ th classical moment [2, 3]. Bettin and Drappeau subsequently studied the large-moment growth rate and answered a question from that work [4]. The present $(2m - 1) \times (2m - 1)$ Fourier model treats arbitrary phases; at the classical phase, reflection reduces the relevant space back to dimension m .

Status	Content
Established background	Classical moment recurrences; the Fabius sinc product; order-two phase analyticity; the cited measure–pressure identity.
Proved in this article	All-positive-integer phase analyticity and a uniform finite-size spectral expansion; the missing $2m$ th Taylor coefficient; the exact global sixth-moment phase optimum.
Additional structural results	The entire spectrum at the atomic phase; a phase-independent determinant; exact arbitrary-phase recurrences and algebraicity statements.
Not established here	Real-analyticity for every noninteger positive order; large- m optimal-phase asymptotics; independent priority or formal proof assistant verification.

The literature and repository search did not locate the all-integer regularity theorem, the all-order missing-coefficient identity, or the sixth-moment optimum in the forms proved below. This is a bounded novelty assessment, not a guarantee that no equivalent result exists elsewhere. The appropriate claim is a proposed research contribution with explicit proofs and reproducible certificates, rather than an unqualified assertion of a major breakthrough.

2 Conventions and the main theorems

All logarithms are natural. We identify $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ and put $e(x) = \exp(2\pi i x)$. Write $\mathcal{D}_n$ for the 2^n dyadic intervals of length 2^{-n} , using closed intervals when taking suprema. Shared endpoints do not affect the integrals.

For an integer $m \geq 1$, define

$$F_{m,n,c}(x) = \prod_{j=0}^{n-1} g_c(T^j x)^m, \quad (2.1)$$

$$Z_{m,n}(c) = \sum_{I \in \mathcal{D}_n} \sup_{x \in I} F_{m,n,c}(x), \quad (2.2)$$

$$p_m(c) = \lim_{n \rightarrow \infty} \frac{1}{n} \log Z_{m,n}(c), \quad (2.3)$$

where existence of the last limit will be proved. This definition agrees with positive-order topological pressure in the conventions of [1]. No use is made of the source’s order-zero normalization: with natural logarithms, the zero-potential pressure is $\log 2$.

The phase-deformed finite product and its moments are

$$Q_{n,c}(x) = \prod_{j=0}^{n-1} (1 + e(2^j x - c)), \quad M_{m,n}(c) = \int_0^1 |Q_{n,c}(x)|^{2m} dx. \quad (2.4)$$

At $c = 1/2$ this is the usual minus-sign Thue–Morse product. The elementary normalization

$$|Q_{n,c}(x)|^{2m} = 4^{mn} F_{m,n,c}(x) \quad (2.5)$$

will account for every power of two below.

Theorem 2.1 (All-integer phase regularity and uniform moment asymptotics). *For every integer $m \geq 1$, there is an explicit matrix $A_m(c)$ of order $2m - 1$ with a positive, algebraically simple eigenvalue $\rho_m(c)$ strictly larger in modulus than every other eigenvalue. Moreover:*

- (i) $p_m(c) = \log \rho_m(c)$ exists and is real-analytic, even, and one-periodic in c .
- (ii) $\text{spec } A_m(0) = \{2^{-j} : 0 \leq j \leq 2m - 2\}$ and $\det A_m(c) = 2^{-(m-1)(2m-1)}$ for every c .
- (iii) There are a positive real-analytic function b_m and constants $K_m < \infty$, $0 < \eta_m < 1$ such that, uniformly in c and $n \geq 0$,

$$M_{m,n}(c) = b_m(c)(2^{2m-1}\rho_m(c))^n(1 + \varepsilon_{m,n}(c)), \quad |\varepsilon_{m,n}(c)| \leq K_m \eta_m^n.$$

The constants are allowed to depend on m .

Theorem 2.2 (The missing Taylor coefficient). For $m \geq 1$, let

$$R_m = \frac{2(2^{2m} - 1)\zeta(2m)}{\pi^{2m}}.$$

As $c \rightarrow 0$, with $t = \pi c$,

$$\rho_m(c) = \cos^{2m} t + R_m t^{2m} + O_m(t^{2m+2}), \quad (2.6)$$

$$p_m(c) = 2m \log \cos t + R_m t^{2m} + O_m(t^{2m+2}). \quad (2.7)$$

In particular,

$$[c^{2m}]p_m(c) = 0, \quad p_m(c) = -2m \sum_{k=1}^{m-1} \frac{(2^{2k} - 1)\zeta(2k)}{k} c^{2k} + O_m(c^{2m+2}). \quad (2.8)$$

For $m = 1$, p_1 is identically zero.

Theorem 2.3 (The fourth and sixth moments have different optimal phases). The unique minimizer of $\rho_2(c)$ modulo one is $c = 1/2$, with

$$\rho_2(1/2) = \frac{1 + \sqrt{17}}{8}.$$

In contrast, $c = 1/2$ is a strict local maximum of $\rho_3(c)$. There are exactly two global minimizers of ρ_3 modulo one:

$$c_* = \frac{\arccos C_*}{2\pi}, \quad 1 - c_*,$$

where C_* is the unique root in $(-0.660930941723, -0.660930941722)$ of the explicit degree-fourteen polynomial in equation (9.8). Numerically,

$$\begin{aligned} c_* &\approx 0.364919195551255, \\ \rho_3(c_*) &\approx 0.430250310745843, \\ 32\rho_3(c_*) &\approx 13.76800994386699. \end{aligned}$$

The two other phases $1/3$ and $1/2$ already give the exact separation

$$\rho_3(1/3) < \frac{7}{16} < \rho_3(1/2). \quad (2.9)$$

The decimal for C_* is certified by a rational interval; the additional numerical displays are approximations, not the basis of the proof.

3 The exact finite transfer model

3.1 An invariant Fourier space

Define the transfer operator using a sum, rather than an average:

$$(\mathcal{L}_{m,c}f)(x) = \sum_{Ty=x} g_c(y)^m f(y). \quad (3.1)$$

Let

$$I_m = \{1 - m, \dots, m - 1\}, \quad \mathcal{E}_m = \text{span}_{\mathbb{C}}\{e(rx) : r \in I_m\}.$$

The real subspace consists of those Fourier coefficients satisfying $v_{-r} = \overline{v_r}$.

The binomial expansion gives

$$g_c(x)^m = \sum_{j=-m}^m a_j e(jx - jc), \quad a_j = 4^{-m} \binom{2m}{m+j}. \quad (3.2)$$

We set $a_j = 0$ outside $[-m, m]$. Since

$$\sum_{Ty=x} e(ky) = \begin{cases} 2e(kx/2), & k \text{ even}, \\ 0, & k \text{ odd}, \end{cases}$$

we obtain the following exact representation.

Proposition 3.1 (Fourier matrix). *The space $\mathcal{E}_m$ is invariant under $\mathcal{L}_{m,c}$. In the basis indexed by I_m , its matrix is*

$$A_m(c)_{k,r} = 2a_{2k-r} e(-(2k-r)c), \quad k, r \in I_m. \quad (3.3)$$

Proof. The product of a degree- m trigonometric polynomial and one of degree at most $m - 1$ has degree at most $2m - 1$. Summing over inverse branches removes odd Fourier frequencies and halves the even ones. The resulting degree is at most $m - 1$. Applying this observation to a basis function and using equation (3.2) gives equation (3.3). $\square$

For example, with $z = e(c)$ and the order $-1, 0, 1$,

$$A_2(c) = \begin{pmatrix} z/2 & z^2/8 & 0 \\ 1/(2z) & 3/4 & z/2 \\ 0 & 1/(8z^2) & 1/(2z) \end{pmatrix}. \quad (3.4)$$

The entries of this matrix need not be real or nonnegative. Positivity below always means positivity of functions, not entrywise positivity of the Fourier matrix.

3.2 Exact moment recovery

Let $v^{(0)}$ denote the coordinate vector of the constant function one, and let ℓ_0 extract the zeroth Fourier coefficient. Iterating the transfer operator and changing variables gives

$$(\mathcal{L}_{m,c}^n 1)(x) = \sum_{T^n y=x} F_{m,n,c}(y), \quad (3.5)$$

$$\ell_0 A_m(c)^n v^{(0)} = 2^n \int_0^1 F_{m,n,c}(x) dx. \quad (3.6)$$

Combining with equation (2.5) yields

$$M_{m,n}(c) = 2^{(2m-1)n} \ell_0 A_m(c)^n v^{(0)}. \quad (3.7)$$

This identity holds for every $n \geq 0$ and every phase, without an asymptotic approximation. In particular, $m = 1$ gives $A_1(c) = (1)$ and $M_{1,n}(c) = 2^n$, independently of c .

4 Simplicity away from the atomic phase

4.1 The positive cone

On the real subspace of $\mathcal{E}_m$, use the cone

$$\mathcal{K}_m = \{f \in \mathcal{E}_m : f(x) \geq 0 \text{ for all } x \in \mathbb{T}\}.$$

It is closed, pointed, generating, and has nonempty interior: the strictly positive functions form its interior. For example, adding a sufficiently large constant expresses any real trigonometric polynomial as a difference of two members of the cone. The operator $\mathcal{L}_{m,c}$ preserves this cone.

We use the elementary finite-dimensional cone version of the Perron principle: a linear map preserving a closed, pointed, solid cone has an eigenvector in the cone at its spectral radius. One justification is to add ϵ times a rank-one strongly positive map. Brouwer's fixed point theorem on a compact normalized cone base gives a positive eigenvector for the perturbed map; its eigenvalue dominates all others by the order-unit norm. Letting $\epsilon \downarrow 0$ and using continuity of the finite spectrum gives the stated principle.

For the two inverse branches, the values of g_c are a and $1 - a$, where $0 \leq a \leq 1$. Therefore

$$2^{1-m} \leq \mathcal{L}_{m,c} 1 = a^m + (1-a)^m \leq 1. \quad (4.1)$$

Iteration and the spectral-radius formula imply

$$2^{1-m} \leq \rho_m(c) \leq 1. \quad (4.2)$$

In particular, the cone eigenvalue is positive.

4.2 A one-zero backward-tree lemma

The weight g_c^m vanishes at exactly one point, $s = c + 1/2 \pmod{1}$. This gives a short replacement for an inappropriate entrywise-matrix irreducibility argument.

Lemma 4.1 (Finite backward-invariant sets). *Suppose that $Z \subset \mathbb{T}$ is nonempty and finite and*

$$T^{-1}Z \setminus \{s\} \subseteq Z.$$

Then $Z = \{0\}$ and $s = 1/2$.

Proof. Writing $r = |Z|$, the left side has at least $2r - 1$ points, so $2r - 1 \leq r$ and $r = 1$. If $Z = \{x\}$, one inverse image of x must be x itself, and the other must be the excluded point s . Thus $Tx = x$. The only fixed point of doubling on $\mathbb{T}$ is zero; its two inverse images are 0 and $1/2$. $\square$

Proposition 4.2 (Strictly positive Perron function). *If $c \neq 0 \pmod{1}$, every nonzero cone eigenfunction $\mathcal{L}_{m,c}h = \rho_m(c)h$ is strictly positive.*

Proof. A nonzero trigonometric polynomial has only finitely many zeros on the circle. If Z is the zero set of h , nonnegativity in the eigenvalue equation implies $T^{-1}Z \setminus \{s\} \subseteq Z$. A nonempty Z would force $s = 1/2$ by lemma 4.1, hence $c = 0$, a contradiction. $\square$

4.3 Excluding other peripheral eigenvalues

Proposition 4.3 (Simple dominant eigenvalue). *For $c \neq 0$, the eigenvalue $\rho_m(c)$ is algebraically simple and is the only eigenvalue of its modulus.*

Proof. Let $h > 0$ be a Perron eigenfunction and let $\mathcal{L}f = \lambda f$, where $|\lambda| = \rho = \rho_m(c)$. Put

$$R = \max_{x \in \mathbb{T}} \frac{|f(x)|}{h(x)}.$$

If $f \neq 0$, then $R > 0$. The function $H = R^2 h^2 - |f|^2$ is a nonnegative trigonometric polynomial. Whenever $H(x) = 0$, equality must hold in

$$|\lambda f(x)| \leq \sum_{Ty=x} g_c(y)^m |f(y)| \leq R \sum_{Ty=x} g_c(y)^m h(y) = R \rho h(x).$$

Consequently, every inverse image of x other than s also lies in the zero set of H . If H were nonzero, its zero set would be finite and nonempty, contradicting lemma 4.1. Hence $|f/h| = R$ everywhere.

Write $F = f/h$. The normalized eigenvalue equation is a convex combination of complex numbers of constant modulus. Equality in the triangle inequality forces

$$F(y) = \frac{\lambda}{\rho} F(Ty) \quad \text{for } y \neq s.$$

Continuity extends this identity to s . At the fixed point zero, $F(0) \neq 0$ gives $\lambda = \rho$. It then follows that $F = F \circ T$. Every dyadic inverse image of zero has the same value as zero; these points are dense, so F is constant. The eigenspace is therefore one-dimensional and no other peripheral eigenvalue exists.

Finally,

$$\|f\|_h = \max_x |f(x)|/h(x) \implies \|\rho^{-1} \mathcal{L}f\|_h \leq \|f\|_h.$$

A nontrivial Jordan block at ρ would cause unbounded polynomial growth of the powers of $\rho^{-1} \mathcal{L}$, which this estimate forbids. Thus the eigenvalue is algebraically simple. $\square$

5 The atomic phase: spectrum and an exact Perron function

The backward-tree argument correctly identifies $c = 0$ as exceptional. Rather than treating this as an ignorable endpoint, we diagonalize its spectrum and identify the positive Perron function explicitly.

5.1 A binomial finite-difference diagonalization

Theorem 5.1 (Complete base-phase spectrum). *For every $m \geq 1$,*

$$\text{spec } A_m(0) = \{1, 2^{-1}, 2^{-2}, \dots, 2^{-(2m-2)}\},$$

and all eigenvalues are simple.

Proof. Let J be a centered binomial random variable with $\Pr(J = j) = a_j$, $-m \leq j \leq m$. Its two parity classes each have probability $1/2$, as follows from $(1 - 1)^{2m} = 0$. For a function P on I_m , equation (3.3) gives

$$(A_m(0)^T P)(r) = \mathbb{E} \left[P \left(\frac{r+J}{2} \right) \middle| J \equiv r \pmod{2} \right]. \quad (5.1)$$

Every admissible $(r + J)/2$ belongs to I_m .

For every polynomial U of degree less than $2m$,

$$\sum_{j=-m}^m (-1)^j a_j U(j) = 0,$$

because this is a constant multiple of a $2m$ th finite difference. Therefore the conditional moments of J of degrees at most $2m - 2$ are equal to the unconditional moments. If $P(r) = r^k$ for $0 \leq k \leq 2m - 2$, the right side of equation (5.1) is a polynomial in r of degree k with leading coefficient 2^{-k} .

The evaluations of $1, r, \dots, r^{2m-2}$ form a basis of functions on I_m by the Vandermonde determinant. In this basis the transpose matrix is triangular with diagonal 2^{-k} . The diagonal values are distinct, proving the theorem. $\square$

5.2 The periodized-sinc eigenfunction

We use $\text{sinc } u = \sin u / u$, with removable value one at zero.

Theorem 5.2 (The base-phase Perron function). *The unique eigenfunction at eigenvalue one normalized by $H_m(0) = 1$ is*

$$H_m(x) = \sum_{k \in \mathbb{Z}} \text{sinc}^{2m}(\pi(x + k)). \quad (5.2)$$

It belongs to $\mathcal{E}_m$ and is strictly positive on $\mathbb{T}$. Equivalently,

$$H_m(x) = \frac{1}{(2m-1)!} \sum_{r=1-m}^{m-1} \left\langle \begin{matrix} 2m-1 \\ m-1+r \end{matrix} \right\rangle e(rx), \quad (5.3)$$

where the brackets denote Eulerian numbers.

Proof. The series in equation (5.2) converges uniformly on a period: its tails are bounded by a constant times $\sum_{|k| \geq K} |k|^{-2m}$. It is positive away from integers because every nonzero term is positive, and equals one at integers.

For completeness, its finite Fourier expansion follows from the Fourier transform of $\text{sinc}^{2m}(\pi x)$. The transform of $\text{sinc}(\pi x)$ is the indicator of $[-1/2, 1/2]$ under the $e^{-2\pi i x \xi}$ convention. Thus the transform of the $2m$ th power is the density B_{2m} of the sum of $2m$ independent uniforms on $[-1/2, 1/2]$. One may justify the product formula first in L^2 and then by convolution, since the even power is integrable. The density has support $[-m, m]$ and vanishes at its endpoints. Periodization therefore has no Fourier frequencies outside I_m .

At an integer r in that interval, the elementary inclusion–exclusion formula for the volume of a clipped simplex gives

$$B_{2m}(r) = \frac{1}{(2m-1)!} \sum_{j=0}^{m+r} (-1)^j \binom{2m}{j} (m+r-j)^{2m-1} = \frac{1}{(2m-1)!} \left\langle \begin{matrix} 2m-1 \\ m-1+r \end{matrix} \right\rangle.$$

The Fourier series of the periodization is absolutely finite, giving equation (5.3). Alternatively, its coefficients follow by integrating the uniformly convergent periodized series over one period.

It remains to verify the eigenvalue equation, which can be done without Fourier transforms. For noninteger x ,

$$H_m(x) = \frac{\sin^{2m} \pi x}{\pi^{2m}} \sum_{k \in \mathbb{Z}} (x+k)^{-2m}.$$

In each term of $\mathcal{L}_{m,0} H_m$, use $\sin \pi y \cos \pi y = \frac{1}{2} \sin 2\pi y$. The branches $y = x/2$ and $y = (x+1)/2$ yield the even and odd terms of the displayed sum after rescaling. Their sum is exactly $H_m(x)$. Continuity gives the identity at integers. Uniqueness follows from theorem 5.1. $\square$

Examples are

$$H_1 = 1, \quad H_2(x) = \frac{2 + \cos 2\pi x}{3}, \quad H_3(x) = \frac{66 + 52 \cos 2\pi x + 2 \cos 4\pi x}{120}.$$

In particular,

$$H_m(1/2) = \frac{1}{\pi^{2m}} \sum_{k \in \mathbb{Z}} (k + 1/2)^{-2m} = \frac{2(2^{2m} - 1)\zeta(2m)}{\pi^{2m}} = R_m. \quad (5.4)$$

The sum of the coefficients in equation (5.3) is one, as it must be from the normalization at zero.

6 Analyticity, pressure identification, and uniform errors

6.1 Analytic perturbation in finite dimensions

Proposition 6.1 (Global analytic Perron data). *For fixed m , the functions $\rho_m(c)$ and $p_m(c) = \log \rho_m(c)$ are real-analytic once pressure is identified below. There are also a real-analytic rank-one spectral projector $P_m(c)$ and a real-analytic positive eigenfunction $h_{m,c}$ normalized by $h_{m,c}(0) = 1$.*

Proof. Every entry of $A_m(c)$ is real-analytic in c . Proposition 4.3 and theorem 5.1 show that the positive dominant eigenvalue is algebraically simple at every real phase. The implicit function theorem applied to the characteristic polynomial therefore gives a local analytic eigenvalue. The locally selected branches agree on overlaps because the positive dominant eigenvalue is unique. The rank-one projector can equivalently be obtained by integrating $(zI - A_m(c))^{-1}$ around a small circle enclosing that eigenvalue.

The projector is positive as an operator on functions: the spectral gap implies $\rho_m(c)^{-n} \mathcal{L}_{m,c}^n \rightarrow P_m(c)$, and each normalized power is positive. If $h > 0$ is any positive Perron function, then $1 \geq h/\|h\|_\infty$, so $P_m(c)1 \geq h/\|h\|_\infty > 0$. Thus

$$h_{m,c} = \frac{P_m(c)1}{(P_m(c)1)(0)}$$

is a well-defined positive analytic normalization, including at $c = 0$ by theorem 5.2. Reflection $x \mapsto -x$ conjugates the phase c operator to the phase $-c$ operator. Hence

$$\rho_m(-c) = \rho_m(c), \quad h_{m,-c}(x) = h_{m,c}(-x). \quad (6.1)$$

Periodicity is immediate from the weights. □

Proposition 6.2 (Uniform spectral estimate). *For fixed m there are constants $K < \infty$ and $0 < \eta < 1$ such that*

$$\|\rho_m(c)^{-n} A_m(c)^n - P_m(c)\| \leq K\eta^n$$

for every real c and every $n \geq 0$, in any fixed matrix norm.

Proof. Set $B(c) = A_m(c)/\rho_m(c)$ and $N(c) = B(c) - P_m(c)$. Every eigenvalue of $N(c)$ has modulus strictly less than one. Continuity of the finite spectrum and compactness of the phase circle give $r_0 = \max_c \rho(N(c)) < 1$. Choose $r_0 < \eta < 1$. The resolvent $(zI - N(c))^{-1}$ is uniformly bounded for $|z| = \eta$ and all c , again by compactness. Cauchy's integral formula for matrix powers gives $\|N(c)^n\| \leq K\eta^n$. Since $P_m N = N P_m = 0$ and $P_m^2 = P_m$, this gives the result for $n \geq 1$; enlarge K to include $n = 0$. □

Applying ℓ_0 and $v^{(0)}$ to this estimate and using equation (3.7) proves the moment expansion in theorem 2.1, with

$$b_m(c) = \ell_0 P_m(c) v^{(0)} > 0. \quad (6.2)$$

The positivity proof in proposition 6.1 and compactness show that b_m is bounded away from zero, so the absolute error may be expressed as a uniform relative error.

At $c = 0$, evaluation at zero is a left eigenfunctional because $(\mathcal{L}_{m,0}f)(0) = f(0)$. Consequently

$$P_m(0)f = f(0)H_m, \quad b_m(0) = \int_0^1 H_m(x) dx = \frac{1}{(2m-1)!} \left\langle \frac{2m-1}{m-1} \right\rangle. \quad (6.3)$$

6.2 Why the finite eigenvalue is the actual pressure

The following comparison avoids an unproved claim that pressure must equal the spectral radius of an arbitrarily chosen invariant subspace.

Lemma 6.3 (Cylinder suprema versus the integral). *If $F \geq 0$ is a trigonometric polynomial of degree at most D and $N = 2^n$, then*

$$N \int_0^1 F \leq \sum_{I \in \mathcal{D}_n} \sup_I F \leq N \int_0^1 F + \int_0^1 |F'| \leq (N + 4\pi D) \int_0^1 F. \quad (6.4)$$

Proof. On an interval I of length $1/N$, the supremum is at least its average and at most its average plus the variation of F on I . Summing gives the first two inequalities.

For the last inequality, factor $F(x) = |P(e(x))|^2$, with P an ordinary polynomial of degree at most D . This is the elementary Fejér–Riesz factorization: reciprocal-conjugate roots of the Laurent polynomial are paired, and roots on the unit circle have even multiplicity because $F \geq 0$; choosing one from each pair gives P . Writing the same function as $P(x) = \sum_{j=0}^D b_j e(jx)$, Parseval and Cauchy–Schwarz give

$$\int |F'| \leq 2\|P'\|_2\|P\|_2 \leq 4\pi D\|P\|_2^2 = 4\pi D \int F.$$

□

The polynomial $F_{m,n,c}$ has degree at most $m(2^n - 1)$. By equations (3.6) and (6.4),

$$\ell_0 A_m(c)^n v^{(0)} \leq Z_{m,n}(c) \leq (1 + 4\pi m) \ell_0 A_m(c)^n v^{(0)}. \quad (6.5)$$

The positive leading coefficient equation (6.2) and the spectral estimate now show that the logarithmic growth of both sides is $\log \rho_m(c)$. This proves existence in equation (2.3) and identifies $p_m(c)$ with the finite eigenvalue. The proof of theorem 2.1(i) and (iii) is complete.

6.3 Strictness away from the atomic phase

Proposition 6.4. *For $m > 1$, $\rho_m(c) = 1$ if and only if $c \equiv 0 \pmod{1}$.*

Proof. We know $\rho_m(0) = 1$. Suppose $\rho_m(c) = 1$ and take a positive Perron function h . At any maximizer x of h , equality in

$$h(x) = \sum_{Ty=x} g_c(y)^m h(y) \leq \|h\|_\infty \sum_{Ty=x} g_c(y)^m \leq \|h\|_\infty$$

forces $a^m + (1-a)^m = 1$. For $m > 1$ this requires $a \in \{0, 1\}$. The two inverse images are therefore c and $c + 1/2$, and $x = 2c$. The branch at c has weight one, so $h(c) = \|h\|_\infty$ as well. Every maximum must equal $2c$, hence $c = 2c$ on the circle and $c = 0$. □

7 Algebraic structure: determinants and recurrences

7.1 A phase-independent determinant

Let $z = e(c)$ and write $A_m(z)$ for the Laurent-polynomial matrix. From equation (3.3),

$$A_m(z) = \text{diag}(z^{-2k})_{k \in I_m} A_m(1) \text{diag}(z^r)_{r \in I_m}.$$

Because $\sum_{k \in I_m} k = 0$,

$$\det A_m(z) = \det A_m(1) = \prod_{j=0}^{2m-2} 2^{-j} = 2^{-(m-1)(2m-1)}. \quad (7.1)$$

This proves theorem 2.1(ii). In particular, none of the finite-model eigenvalues can pass through zero as the phase changes.

7.2 Characteristic polynomials in a single real parameter

Put $d = 2m - 1$ and $C = \cos 2\pi c$. The principal-minor expansion gives

$$\chi_m(\lambda, z) = \sum_{S \subseteq I_m} (-1)^{|S|} \lambda^{d-|S|} \det A_m(1)[S, S] z^{-\sum_{k \in S} k}. \quad (7.2)$$

Reflection $k \mapsto -k$ leaves the relevant determinants unchanged, so this Laurent polynomial is invariant under $z \mapsto z^{-1}$. It is therefore a polynomial in $C = (z + z^{-1})/2$ with rational coefficients. The largest possible absolute subset sum in equation (7.2) is $m(m-1)/2$, proving

$$\chi_m(\lambda, C) \in \mathbb{Q}[\lambda, C], \quad \deg_C \chi_m \leq \frac{m(m-1)}{2}. \quad (7.3)$$

To distinguish phase and cosine parameters, set

$$\varrho_m(C) = \rho_m\left(\frac{\arccos C}{2\pi}\right).$$

This is an analytic function of C in a neighborhood of each point of $[-1, 1]$. Indeed, it is a simple root of χ_m and the real implicit function theorem applies directly in (λ, C) , including at the two endpoints.

Corollary 7.1 (Algebraicity at arithmetic phases). *If $c = a/b$ is rational, then $\rho_m(c)$ and $2^{2m-1}\rho_m(c)$ are algebraic numbers of degree at most $(2m-1)\varphi(b)$ over $\mathbb{Q}$. More generally, they are algebraic whenever $\cos 2\pi c$ is algebraic.*

Proof. At a rational phase, $z = e(a/b)$ lies in a cyclotomic field of degree at most $\varphi(b)$. The eigenvalue satisfies the degree- $2m-1$ characteristic polynomial over that field. The second assertion follows from equation (7.3). $\square$

7.3 Exact linear recurrences

Write $\chi_m(\lambda, c) = \lambda^d + \gamma_1(c)\lambda^{d-1} + \cdots + \gamma_d(c)$, and set $u_n(c) = \ell_0 A_m(c)^n v^{(0)}$. Cayley–Hamilton yields

$$u_{n+d} + \gamma_1 u_{n+d-1} + \cdots + \gamma_d u_n = 0 \quad (n \geq 0). \quad (7.4)$$

Since $M_{m,n} = 2^{(2m-1)n} u_n$, this immediately gives a recurrence for the actual moments as well. The stated order is an upper bound; symmetry or cancellation can reduce the minimal order.

At $c = 0$ and $c = 1/2$, the weight is even. Starting from the constant function, the operator stays in the even trigonometric subspace $\text{span}\{1, \cos 2\pi x, \dots, \cos 2\pi(m-1)x\}$, of dimension m . This explains compatibility with the earlier classical order- m recurrence [2, 3]; the larger arbitrary-phase matrix is not claimed to improve that known bound.

8 Delayed feedback and the missing coefficient

8.1 An exact equation at the fixed point

Use the positive analytic normalization $h_{m,c}(0) = 1$ from proposition 6.1. Evaluating the transfer equation at zero exposes its two inverse branches:

$$\rho_m(c) = \cos^{2m}(\pi c) + \sin^{2m}(\pi c) h_{m,c}(1/2). \quad (8.1)$$

The first summand comes from the fixed point zero. The second is the feedback from the other inverse branch; its weight has a zero of order $2m$ at the atomic phase.

Reflection equation (6.1), together with $-1/2 = 1/2$ on the circle, gives

$$h_{m,-c}(1/2) = h_{m,c}(1/2).$$

Thus this value is an even analytic function of c . By equation (5.4),

$$h_{m,c}(1/2) = R_m + O_m(c^2). \quad (8.2)$$

Substituting into equation (8.1), and using $\sin^{2m} t = t^{2m} + O_m(t^{2m+2})$, proves equation (2.6).

To pass to logarithms, factor out $\cos^{2m} t$ near zero. The quotient is $1 + R_m t^{2m} + O_m(t^{2m+2})$. Since its square-error term is $O(t^{4m}) \subseteq O(t^{2m+2})$ for every $m \geq 1$, this proves equation (2.7).

8.2 Why the coefficient cancels exactly

The Euler product for cosine is

$$\cos t = \prod_{j=0}^{\infty} \left(1 - \frac{4t^2}{\pi^2(2j+1)^2} \right).$$

It follows from the sine product by dividing $\sin 2t/(2t)$ by $\sin t/t$, or by pairing its zeros and using its value at zero. For $|t| < \pi/2$, absolute convergence permits logarithmic expansion:

$$\log \cos t = - \sum_{k \geq 1} \frac{(2^{2k} - 1)\zeta(2k)}{k\pi^{2k}} t^{2k}. \quad (8.3)$$

The coefficient of t^{2m} in $2m \log \cos t$ is

$$- \frac{2(2^{2m} - 1)\zeta(2m)}{\pi^{2m}} = -R_m.$$

This is precisely the negative of the feedback coefficient in equation (2.7). The degree- $2m$ term vanishes, proving theorem 2.2.

This is not a claim that the whole pressure is flat to order $2m$. For $m > 1$, its leading curvature is

$$p_m''(0) = -2m\pi^2 < 0. \quad (8.4)$$

The cancellation removes one specific higher even coefficient, while all lower even coefficients agree with the fixed-point contribution.

8.3 Exact low-order expansions

Let $t = \pi c$. The checker in the accompanying package computes the following Taylor coefficients using exact rational arithmetic:

$$p_2(c) = -2t^2 + \frac{376}{45}t^6 + O(t^8), \quad (8.5)$$

$$p_3(c) = -3t^2 - \frac{1}{2}t^4 + \frac{213}{28}t^8 + O(t^{10}), \quad (8.6)$$

$$p_4(c) = -4t^2 - \frac{2}{3}t^4 - \frac{8}{45}t^6 + \frac{16978624}{3075975}t^{10} + O(t^{12}). \quad (8.7)$$

The all-order theorem proves the missing terms t^4, t^6, t^8 in these three displays. The next nonzero coefficients are finite exact calculations rather than an asserted all-order pattern. For example, $376/45$ in equation (8.5) is not inferred from the sign of a numerical plot.

m	2	3	4	5	6
R_m	1/3	2/15	17/315	62/2835	1382/155925
Missing degree in p_m	4	6	8	10	12

9 Fourth- and sixth-moment phase optimization

It is enough to minimize $\rho_m(c)$ because the exponential moment rate is $2^{2m-1}\rho_m(c)$, and the multiplier is phase-independent. By evenness and periodicity, we may work in $C = \cos 2\pi c \in [-1, 1]$. The characteristic-polynomial reductions used in this section are checked directly from equation (3.3) in the supplied scripts.

9.1 The fourth moment: a unique classical optimum

For $m = 2$,

$$\chi_2(\lambda, C) = \lambda^3 - (3/4 + C)\lambda^2 + (1/4 + 5C/8)\lambda - 1/8. \quad (9.1)$$

At every C ,

$$\chi_2(5/8, C) = -9/512 < 0.$$

Since the polynomial is positive for sufficiently large λ , there is a positive root greater than $5/8$. The Perron root therefore satisfies $\varrho_2(C) > 5/8$.

At the Perron root, $\partial_\lambda \chi_2 > 0$. More generally, for a real monic polynomial with a simple strictly dominant positive root, every other real root contributes a positive difference, and complex-conjugate roots contribute positive squared moduli. Hence implicit differentiation gives

$$\frac{d\varrho_2}{dC} = -\frac{\varrho_2(5/8 - \varrho_2)}{\partial_\lambda \chi_2(\varrho_2, C)} > 0.$$

Thus the minimum occurs uniquely at $C = -1$. At that phase,

$$\chi_2(\lambda, -1) = (\lambda + 1/2)(\lambda^2 - \lambda/4 - 1/4),$$

and the positive root is $(1 + \sqrt{17})/8$. This proves the fourth-moment assertion in theorem 2.3.

9.2 The sixth-moment characteristic polynomial

For $m = 3$, it is convenient to clear denominators and write $f(\lambda, C) = 2048\chi_3(\lambda, C)$:

$$\begin{aligned} f(\lambda, C) = & 2048\lambda^5 - 128(12C^2 + 15C + 4)\lambda^4 \\ & + 40(28C^3 + 24C^2 + 9C + 1)\lambda^3 \\ & - (448C^3 + 261C^2 - 126C + 37)\lambda^2 \\ & + (63C^2 + 21C - 22)\lambda - 2. \end{aligned} \quad (9.2)$$

At the classical phase,

$$f(\lambda, -1) = 2(16\lambda^2 + 4\lambda - 1)(64\lambda^3 - 20\lambda^2 - 6\lambda + 1). \quad (9.3)$$

At phase $c = 1/3$, where $C = -1/2$,

$$4f(\lambda, -1/2) = 8192\lambda^5 + 256\lambda^4 - 160\lambda^3 - 437\lambda^2 - 67\lambda - 8 =: P(\lambda). \quad (9.4)$$

There is exactly one positive root of P by Descartes' rule of signs. Since $P(0) < 0$ and

$$P(7/16) = 405/64 > 0, \quad (64\lambda^3 - 20\lambda^2 - 6\lambda + 1)_{\lambda=7/16} = -3/32 < 0, \quad (9.5)$$

we get the exact strict separation $\rho_3(1/3) < 7/16 < \rho_3(1/2)$. This alone disproves the natural extrapolation that the classical phase minimizes every positive even moment.

9.3 The classical phase is a strict local maximum

At $C = -1$, the Perron root is greater than $7/16$. The positive root of the quadratic factor in equation (9.3) is less than $1/4$, so the Perron root belongs to the cubic factor. Reduction modulo that cubic gives

$$f_C(\rho, -1) = 3\rho(720\rho^2 - 196\rho - 41), \quad \rho = \rho_3(1/2). \quad (9.6)$$

The quadratic in parentheses is positive for $\rho \geq 7/16$: its value at $7/16$ is $177/16$, and its derivative is positive there and thereafter. Also $f_\lambda(\rho, -1) > 0$ by the simple dominant root property. Thus $d\rho_3/dC < 0$ at $C = -1$.

Writing $c = 1/2 + u$,

$$C = -1 + 2\pi^2 u^2 + O(u^4).$$

It follows that $\rho_3(1/2 + u) < \rho_3(1/2)$ for sufficiently small nonzero u . The same is true of p_3 and of the sixth-moment exponential growth factor.

9.4 Certifying the global minimizer

A stationary Perron root in $-1 < C < 1$ satisfies

$$f(\rho, C) = 0, \quad f_C(\rho, C) = 0,$$

because $f_\lambda(\rho, C) \neq 0$. Eliminate ρ :

$$\text{Res}_\lambda(f, f_C) = -3057647616 R(C), \quad (9.7)$$

where

$$\begin{aligned} R(C) = & 478310547456C^{14} - 263889031168C^{13} - 2211948322304C^{12} \\ & + 2274923686784C^{11} + 2772364425376C^{10} - 3601461331552C^9 \\ & - 2093687468208C^8 + 2695719201426C^7 + 1231497289965C^6 \\ & - 1032213437124C^5 - 514151253800C^4 + 146152959398C^3 \\ & + 104125846753C^2 + 11781400244C + 375091954. \end{aligned} \quad (9.8)$$

A resultant alone is not enough: some of its zeros can represent stationary non-Perron eigenvalues. The following certificate explicitly removes those extraneous candidates.

Let $V(x)$ be the number of sign changes in the Sturm sequence of R at x , ignoring zero entries. Exact rational arithmetic gives $V(-1) = 9$, $V(1) = 5$, and the four isolating intervals below. Every decimal endpoint in this table denotes an exact rational number.

Interval for C	$V(\text{left}), V(\text{right})$	Common-root enclosure
$(-0.660930941723, -0.660930941722)$	9, 8	$(0.430, 0.431)$
$(-0.070099130476, -0.070099130475)$	8, 7	$(-0.152, -0.150)$
$(-0.068679451074, -0.068679451073)$	7, 6	$(-0.134, -0.132)$
$(0.914028755782, 0.914028755783)$	6, 5	$(0.229, 0.231)$

Here is how the last column is certified, rather than numerically assigned. The degree-one subresultant of f and f_C is

$$169869312(A(C)\lambda + B(C)), \quad (9.9)$$

where the integer polynomials A, B are displayed in section A. Subresultants are polynomial combinations of the original two polynomials. The extracted content 169869312 is a nonzero constant, not a parameter-dependent factor. Consequently every common root satisfies $A(C)\lambda + B(C) = 0$.

Rational interval Horner evaluation on each listed interval shows that $A(C)$ is bounded away from zero and that $-B(C)/A(C)$ belongs to the stated enclosure. Thus any stationary eigenvalue corresponding to the last three rows is less than $1/4$. But equation (4.2) gives $\varrho_3(C) \geq 1/4$ for all C , so none of those rows can represent a stationary Perron eigenvalue.

A global Perron minimum exists by continuity. It lies in the interior because the value at $C = -1/2$ is smaller than both endpoint values: $\varrho_3(-1) > 7/16$, $\varrho_3(1) = 1$, and $\varrho_3(-1/2) < 7/16$. Every interior minimum is stationary. Hence the first row must contain a stationary Perron point, and it is the only one. This proves uniqueness of the minimizing C_* and, by reflection, exactly two minimizing phases modulo one.

For orientation, high-precision numerical evaluation gives

$$C_* \approx -0.6609309417223258423, \quad \rho_3(1/2) \approx 0.4443474454062066009.$$

For comparison, the exponential sixth-moment rates are

$$32\rho_3(1/2) \approx 14.21911825299861, \quad 32\rho_3(c_*) \approx 13.76800994386699.$$

The strict inequalities and uniqueness do not depend on these decimals.

10 Consequences for dimensions and the Fabius program

10.1 Integer Rényi dimensions

This subsection uses an external theorem, unlike the finite-operator proofs above. Define the generalized Thue–Morse measure by the weak limit of Riesz-product densities

$$\mu_c = \text{w-lim}_{n \rightarrow \infty} \left(\prod_{j=0}^{n-1} 2g_c(T^j x) \right) dx.$$

The existence and the identity

$$\beta_{\mu_c}(m) = p_m(c)/\log 2 \quad (c \neq 0, m \geq 1) \quad (10.1)$$

are imported from [1, Theorem 2.6]. Here

$$\beta_\mu(q) = \lim_{n \rightarrow \infty} \frac{1}{n \log 2} \log \sum_{I \in \mathcal{D}_n} \mu(I)^q,$$

using a dyadic partition convention for the measure. For an integer $m > 1$, the Rényi dimension is $D_m(\mu) = \beta_\mu(m)/(1 - m)$.

Corollary 10.1 (Analytic integer dimension functions). *With the cited measure–pressure identity, for every integer $m > 1$,*

$$D_m(\mu_c) = \frac{p_m(c)}{(1 - m) \log 2} \quad (10.2)$$

is real-analytic on the entire phase circle. Near $c = 0$,

$$D_m(\mu_c) = \frac{2m}{(m - 1) \log 2} \sum_{k=1}^{m-1} \frac{(2^{2k} - 1)\zeta(2k)}{k} c^{2k} + O_m(c^{2m+2}),$$

and its c^{2m} coefficient is zero.

Proof. For $c \neq 0$ the formula is equation (10.1). At $c = 0$, the measure is δ_0 [1], so every positive-order dyadic mass sum equals one and $D_m = 0$. Also $p_m(0) = 0$ for $m \geq 1$ by our base spectrum. Thus the equality extends at these integer orders. Analyticity and the expansion follow from theorems 2.1 and 2.2. $\square$

The restriction to these orders matters: we have not asserted that the measure–pressure identity extends at the atomic phase for all real orders. Nor does the corollary give the entropy dimension at $m = 1$ by substituting into a formula with zero denominator.

Corollary 10.2 (Distinct dimension-maximizing phases). *Under the same cited identity, $D_2(\mu_c)$ has its unique maximum at $c = 1/2$, whereas $D_3(\mu_c)$ has exactly two global maxima at $c_*, 1 - c_*$. At $c = 1/2$, D_3 has a strict local minimum.*

Proof. The denominator in equation (10.2) is negative. Apply theorem 2.3 and the strict monotonicity of the logarithm. $\square$

Thus a phase optimized for one integer spectral dimension need not be optimized for the next. The sixth-moment computation is not merely an alternative decimal evaluation of the known fourth-moment exponent.

10.2 What the result says about the up-function

Under the Fourier convention $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \xi} dx$, the Rvachev up-function has the classical transform

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^h}\right). \quad (10.3)$$

See [5, 7]. Factoring its first n dilated terms gives for $x \neq 0$

$$\Phi(2^n x) = \Phi(x) (\pi x)^{-n} 2^{-n(n+1)/2} \prod_{j=1}^n \sin(\pi 2^j x). \quad (10.4)$$

The oscillatory factor is exactly

$$2^n \left| \prod_{j=1}^n \sin(\pi 2^j x) \right| = |Q_{n,1/2}(2x)|.$$

Since doubling preserves Lebesgue measure on the circle,

$$\int_0^1 \prod_{j=1}^n |2 \sin(\pi 2^j x)|^{2m} dx = M_{m,n}(1/2). \quad (10.5)$$

Thus the finite model directly computes the exponential rates of all even moments of the stripped Fourier numerator.

The new phase theorems provide a controlled deformation of that numerator and exact comparison phases. They can be used as analytic benchmarks for pressure code, as finite algebraic test cases for formalization, and as obstructions to assuming the classical phase is universally moment-optimal.

There is an important limitation. The factors x^{-n} and $\Phi(x)$ in equation (10.4) are not constant on a shell. Therefore equation (10.5) is not automatically an asymptotic for an unweighted Fourier-shell integral. No new universal pointwise decay exponent, annular maximum formula, or noninteger shell-pressure theorem is claimed. The distinctions between these quantities are already central to the repository's Fourier-decay draft [7].

11 Reproducible computation and a formalization route

11.1 Exact Taylor recursion

The matrices give a rational algorithm for arbitrary finite jets at $c = 0$. Use $t = \pi c$ and write

$$A(t) = \sum_{n \geq 0} A_n t^n, \quad v(t) = \sum_{n \geq 0} v_n t^n, \quad \rho(t) = \sum_{n \geq 0} \lambda_n t^n,$$

where $\lambda_0 = 1$. By equation (3.3),

$$(A_n)_{k,r} = (A_0)_{k,r} \frac{[-2i(2k-r)]^n}{n!}. \quad (11.1)$$

Let u be the row of ones, so uv is evaluation at zero. Normalize $uv_0 = 1$ and $uv_n = 0$ for $n \geq 1$. The vector v_0 is the Eulerian coefficient vector in equation (5.3).

For each $n \geq 1$, solve the same bordered linear system

$$\begin{pmatrix} A_0 - I & -v_0 \\ u & 0 \end{pmatrix} \begin{pmatrix} v_n \\ \lambda_n \end{pmatrix} = \begin{pmatrix} -\sum_{k=1}^n A_k v_{n-k} + \sum_{k=1}^{n-1} \lambda_k v_{n-k} \\ 0 \end{pmatrix}. \quad (11.2)$$

The matrix is invertible because eigenvalue one is simple and $uv_0 = 1$. Its entries lie in $\mathbb{Q}(i)$, so every coefficient is exact. If $p(t) = \log \rho(t) = \sum_{n \geq 1} p_n t^n$, then differentiating $\rho = \exp p$ gives the recursion

$$p_n = \lambda_n - \sum_{k=1}^{n-1} \frac{k}{n} p_k \lambda_{n-k}. \quad (11.3)$$

This is a finite algebraic way to check any desired coefficient without numerically differentiating a nearly degenerate matrix.

11.2 What was actually executed

The package includes two Python checkers requiring SymPy. The execution receipts were generated with SymPy 1.14.0. Neither checker uses floating point arithmetic for its assertions.

`verify_pressure.py`. The default run passes 23 groups of checks. It verifies the base spectrum, Eulerian Perron vector, and determinant for $1 \leq m \leq 10$; symbolic phase-independent determinants and reflection for $1 \leq m \leq 4$; exact Parseval moment identities for $1 \leq m \leq 4$, $0 \leq n \leq 6$, and both signs; and exact Taylor coefficients through degree $2m + 2$ for $2 \leq m \leq 6$. The receipt is `verification.json`.

`verify_minimum.py`. This checks the sixth-moment characteristic polynomial directly from the matrix, computes the exact resultant and linear subresultant, constructs a rational Sturm sequence, proves the four isolating intervals, and bounds each candidate common eigenvalue by rational interval Horner arithmetic. It also checks the separator values at $7/16$. The receipt is `minimum_certificate.json`.

These finite checks are safeguards against transcription, scaling, and low-order algebra mistakes. They do not replace the all- m proofs of analyticity or coefficient cancellation. The global sixth-moment theorem does use the finite exact certificate as part of its proof; the integer polynomials and verification recipe are supplied so it can be audited independently.

11.3 A practical Lean decomposition

No Lean build was performed for this article. A proposed implementation should keep the analytic and finite algebraic components separate.

The first layer would define finite Laurent polynomials and prove the coefficient formula [equation \(3.3\)](#), the moment identity, the phase-independent determinant, and the characteristic polynomials for $m = 2, 3$. These are finite algebraic targets and do not need a general thermodynamic-formalism library.

A second layer would formalize the finite backward-tree lemma and the fact that a nonzero trigonometric polynomial has finitely many zeros. The positivity and peripheral-spectrum proof can then be phrased in terms of functions and a weighted supremum norm. This avoids trying to apply an entrywise-positive-matrix theorem to a complex Fourier matrix.

A third layer would prove the binomial finite-difference spectrum at $c = 0$ and the normalized H_m formula. The finite Eulerian version may be the easier route for exact values, whereas periodization makes the analytic proof of R_m conceptually transparent.

Finally, a spectral perturbation layer would deliver real-analytic Perron data and feed the fixed-point equation [equation \(8.1\)](#) into a power-series argument. The sixth-moment certificate could be imported as explicit polynomial identities and rational sign checks, but a kernel checked Sturm theorem would be preferable to trusting an external computer-algebra receipt.

This order produces useful intermediate theorems even before the entire article is formalized. It also preserves a visible boundary between proof-assistant results, exact external computations, and proposed analytic extensions.

12 Further research questions

The following questions are proposed continuations, not claims that they have all been identified as open in the published literature. The first is the remaining part of the explicit regularity question in [1].

12.1 Noninteger regularity and the atomic threshold

Question 12.1. For a fixed noninteger $q > 0$, what is the exact regularity of $c \mapsto P(q\psi_c)$, especially at $c = 0$?

The integer proof relies on a finite Fourier cutoff that disappears for noninteger q . The fixed-point equation suggests examining whether a term of order $|c|^{2q}$ appears and whether special cancellations can occur. This is a heuristic direction, not an established noninteger expansion. A rigorous route would need a suitable eigenfunction space, control of the singular branch, and a proof that the relevant eigenvalue still represents pressure. Different regimes near $q = 1/2$ deserve separate treatment rather than extrapolating the integer argument.

12.2 The first coefficient after the cancellation

Question 12.2. Is $[t^{2m+2}]p_m(t/\pi)$ positive for every integer $m \geq 2$? Can it be written in a closed form in m ?

The executed checks are positive for $2 \leq m \leq 6$. Differentiating [equation \(8.1\)](#) two additional times reduces the problem to the second phase derivative of $h_{m,c}(1/2)$ at zero. The completely explicit base spectrum might permit a finite resolvent sum with a useful sign structure.

Editorial note (ProveIt, 2026-10-01). The positivity question is answered, and the executed checks are overtaken, by thirteen later packages filed beside this article on 2026-10-01 (batch 72 of the repository's docs/incoming/; unreviewed). `.../Thue_Morse_Integer_Pressure_First_Positive/` proves $[t^{2m+2}]p_m(t/\pi) > 0$ for every integer $m \geq 2$ along the route suggested here: the quadratic coefficient in $a = \tan \pi c$ of the normalized eigenfunction at $1/2$ is a finite sum over the atomic eigenbasis with positive rational coefficients built from Bernoulli numbers, and

$$[t^{2m+2}]p_m(t/\pi) = \sum_{j=1}^{m-1} b_{m,j} R_{m-j} + \frac{2m}{3} R_m - \frac{m}{m+1} R_{m+1},$$

with R_m as in [theorem 2.2](#) and explicit positive rationals $b_{m,j}$. That is a closed form in this finite sense; no elementary closed form is given, and the reduced denominators are left open there. Moreover every even coefficient strictly between degrees $2m$ and $6m$ is positive for every $m \geq 2$ (`.../Thue_Morse_Integer_Pressure_Full_Range/`), and degree $6m$ cannot be included in general: $[t^{12}]p_2(t/\pi) = -35360872/93555$. The first negative degree N_m above $2m$ satisfies $N_m = \gamma m - \Lambda^{-1} \log \log(2m) + O(1)$ with $\gamma \in (6.662966, 6.662968)$ (`.../Thue_Morse_Integer_Pressure_First_Negative/`), and it is certified for $2 \leq m \leq 128$ in `.../Thue_Morse_Integer_Pressure_First_Negative_Data/`. The other packages of the series are listed in the README of `.../`.

12.3 Optimal phases for higher moments

Question 12.3. For each $m \geq 4$, how many phases globally minimize $\rho_m(c)$? How do those phases depend on m ?

The change between $m = 2$ and $m = 3$ rules out a simple universal classical-phase principle. The characteristic polynomial in C gives an exact elimination procedure at each fixed m , but resultant size grows rapidly. A combination of cone bounds, symmetry, and certified interval eigenvalue methods may scale better than full elimination.

12.4 The large-moment phase landscape

Question 12.4. Do minimizing phases converge as $m \rightarrow \infty$? Can their accumulation points be characterized by ergodic optimization of $\log g_c$?

Existing large-moment results at the classical phase [4] should be treated as input and comparison, not rediscovered. A new result would have to control the phase uniformly. In particular, a fixed- m spectral gap cannot simply be interchanged with the limit $m \rightarrow \infty$.

12.5 Minimal recurrences and algebraic degree

Question 12.5. What is the generic minimal recurrence order of $M_{m,n}(c)$, and which phases cause it to drop below $2m - 1$?

Reflection explains the reduction at 0 and $1/2$, but other cancellations could occur between the initial vector, spectral projectors, and the integration functional. The appropriate finite criterion is the rank of the corresponding Hankel or Krylov matrix, not merely the degree of the characteristic polynomial.

12.6 Quantitative gaps and certified errors

Question 12.6. Can the constants K_m, η_m in the uniform moment estimate be bounded explicitly and efficiently in m ?

Compactness proves their existence but is not a numerical certificate. The one-zero argument proves strictness without a ready-made lower bound on separation. An effective proof could use a bounded iterate that contracts the positive cone, with a separate treatment near the atomic phase using its exact spectrum.

12.7 Beyond one zero and beyond base two

Question 12.7. Which nonnegative trigonometric weights for $x \mapsto bx$ admit an analogous all-phase simple finite Perron model?

The degree cutoff generalizes immediately. The subtle part is spectral simplicity. With s forbidden inverse images and a finite equality set of size r , the counting argument becomes $br - s \leq r$. Its possible small invariant sets should determine exceptional parameter loci. This suggests a classification theorem organized by finite backward orbits of the zero set.

Editorial note (ProveIt, 2026-09-30). . . . /Periodic_Anchors_Exact_Mixed_Moment_Phase_Diagrams/ (batch 68 of the repository's docs/incoming/) computes exactly the pressure of products of shifted digital masks whose phases fill complete periodic orbits of $x \mapsto bx \bmod 1$, with a common exponent on each orbit, together with every equilibrium measure and the resulting mixed-moment exponents, in every base. Such a product has several zeros, all mapped by $x \mapsto bx$ onto the selected orbits, and its pressure is a maximum of explicit orbit branches; no finite Perron model or spectral simplicity is proved. It bears on this question but does not answer it: the phases are fixed, a single mask whose phase has period at least two is not covered, and no regularity in the phase is proved.

12.8 Phase sequences and matrix products

Question 12.8. What changes when the phase is allowed to vary with scale, replacing c by c_j in the j th factor?

The exact moment representation becomes a product of the same finite matrices. Periodic phase sequences are finite algebraic models; more general sequences lead to Lyapunov exponents and joint spectral radii. This extension may distinguish optimization by a constant phase from optimization by a scale-dependent protocol.

12.9 A kernel-checked optimization certificate

Question 12.9. Can the sixth-moment optimum be formalized with a small, transparent certificate whose correctness is checked entirely in Lean?

The proof naturally splits into matrix identities, the analytic simple root theorem, and a degree-fourteen real-root certificate. Establishing this example would create a reusable workflow for moment optimization problems elsewhere in ProveIt.

13 Conclusions and scope

The integer powers of the generalized Thue–Morse weight turn a singular pressure problem into finite Fourier algebra. The key spectral step is a one-zero backward-tree argument, supplemented by an exact treatment of the atomic phase. Together they establish real-analytic phase dependence for every positive integer order and a uniform exponential moment expansion.

The fixed-point eigenvalue equation then explains a more unexpected identity: feedback from the second branch enters precisely at degree $2m$ and cancels that coefficient of the pressure. The sixth-moment calculation shows that the phase geometry is genuinely richer than the fourth-moment case: the classical phase ceases to minimize the rate, and the true optimum has an explicit algebraic characterization.

These conclusions concern a concrete slice of the repository’s Fabius/Thue–Morse program. They do not resolve all positive real pressure orders, do not determine a universal Fourier-decay law, and have not been checked by a proof assistant or independent referee. The supplied proofs and exact certificates make those remaining validation tasks specific rather than leaving the central assertions at the level of numerical evidence.

A The sixth-moment certificate polynomials

The polynomials in equation (9.9) are

$$\begin{aligned} A(C) = & -34165039104C^{13} - 10163951616C^{12} + 145543077120C^{11} \\ & - 39584395200C^{10} - 236661608848C^9 + 88141321128C^8 \\ & + 214754071532C^7 - 68497376017C^6 - 115302016540C^5 \\ & + 21790610488C^4 + 33317271568C^3 - 2056400573C^2 \\ & - 4362939336C - 301154570, \end{aligned} \tag{A.1}$$

$$\begin{aligned} B(C) = & 3253813248C^{11} - 2632283136C^{10} - 7946511104C^9 \\ & + 9092577984C^8 + 8363143264C^7 - 10354590704C^6 \\ & - 4894201844C^5 + 5242980644C^4 + 2297293628C^3 \\ & - 801503452C^2 - 418577016C - 27190504. \end{aligned} \tag{A.2}$$

A reader can independently reproduce the certificate as follows. Compute the resultant and polynomial subresultant sequence of equation (9.2) and its C derivative, eliminating λ . Check equations (9.7) and (9.9). Construct the Sturm sequence of R using ordinary Euclidean remainders with reversed sign. Evaluate its signs at $-1, 1$ and the eight rational endpoints in the table in section 9. Finally, interval-evaluate A and B by Horner’s rule, check that the denominator interval excludes zero, and divide the interval for $-B$ by that for A .

Every operation in this recipe is addition, multiplication, comparison, or exact division of rational numbers. Floating-point root finding, plot sampling, and heuristic root matching are unnecessary.

B Build and verification instructions

The source uses a standard $\text{\LaTeX}$ installation with the Libertinus, AMS, microtype, hyperref, and cleveref packages. No repository checkout or network access is needed to compile it. From the package directory:

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
python verify_pressure.py
python verify_minimum.py
```

The last two commands require Python 3.10 or later and SymPy; the tested version is recorded in `requirements.txt`. The algebraic checks write JSON receipts to the working directory. Avoid running Python with `-O`, because assertions are part of these verification scripts. The article is self-contained; its bibliography URLs are for provenance and further reading, not build dependencies.

References

- [1] P. Gohlke, M. Kesseböhmer, and T. I. Schindler, *Generalized Thue–Morse measures: spectral and fractal analysis*, arXiv:2509.22109v1, 26 September 2025. <https://arxiv.org/abs/2509.22109>. Theorem 2.7 and its following paragraph give the regularity question; Theorems 2.3 and 2.6 give the pressure identifications used for context and the measure corollary. Version checked on 28 September 2026.
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- [7] *Fourier Decay of Rvachev’s Up-Function: Exact products, pressure spectra, Cesàro gauges, fluctuations, peaks, and valleys*, research synthesis in the ProveIt snapshot of [6]. Repository path: `Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay/Rvachev_Up_Fourier_Decay.tex`. The inspected source is dated 31 August 2026. It is used as repository context, not as an independently refereed source or an unconditional black-box proof of the present results.